Predictive Phase Report

PETase Engineering Tournament 2025 – Zero-Shot Track

Organized by the Align Foundation

Team Information

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1. Objective

This work presents a fully computational zero-shot framework for predicting enzymatic turnover numbers (kcat) and derived specific activities for PETase variants submitted to the **Protein Engineering Tournament 2025**.

No experimental kinetic measurements from the tournament dataset were used in model calibration, ensuring strict compliance with Zero-Shot Track guidelines.

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2. Molecular Weight Determination from Primary Sequence

2.1 Rationale

Turnover number (kcat) is defined per mole of enzyme, whereas experimentally reported activity is typically expressed as:

$$\text{Specific Activity} = \frac{\mu mol\ product}{min \cdot mg\ enzyme}$$

Therefore, molecular weight (MW) is required to interconvert between molar turnover (kcat) and mass-normalized activity.

2.2 Computational Procedure

Protein sequences in FASTA format were processed using BioPython in Google Colab.

Molecular weight was computed as:

$$MW = \sum_{i=1}^{n} m_i - (n-1) \times 18.015$$

Where:

- m_i= residue mass of amino acid i (Da)
- n= sequence length
- 18.015 Da accounts for water loss during peptide bond formation

Outputs per sequence:

- Protein ID
- Sequence length
- Molecular weight (Da)

This ensures physically accurate molecular mass estimation directly from primary sequence.

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3. Turnover Number (kcat) Prediction

3.1 Model Framework

Turnover numbers were predicted using the machine learning framework described in:

Turnover number predictions for kinetically uncharacterized enzymes using machine and deep learning
Kroll et al., Nature Communications (2023)
DOI: 10.1038/s41467-023-39840-4

This model integrates sequence-derived features and learned representations to estimate kcat for kinetically uncharacterized enzymes.

3.2 Zero-Shot Compliance

- No experimental activity values from the competition dataset were used.
- Predictions were derived solely from sequence-level information.
- No supervised fine-tuning was performed on tournament variants.

Thus, the methodology strictly adheres to Zero-Shot Track requirements.

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4. Conversion from kcat to Specific Activity

4.1 Theoretical Basis

By definition:

$$k_{cat} = \frac{V_{max}}{[E]_{total}}$$

To express activity per mg enzyme:

$$Activity = \frac{k_{cat} \times 60}{MW}$$

Where:

- kcat in s^{-1}
- MW in g/mol (Da)
- 60 converts seconds to minutes

This formula results directly from dimensional analysis and standard enzyme kinetics definitions.

4.2 Computational Implementation

A Python script:

1. Uploads a CSV containing:
   - MW_Da
   - kcat [s^(-1)]

2. Computes:

$$Activity(\mu mol/min \cdot mg) = \frac{kcat \times 60}{MW}$$

3. Outputs a processed CSV containing predicted specific activities.

All computations were performed in Google Colab using pandas for data processing.

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5. Solubility Prediction Using NetSolP

5.1 Rationale

The expression efficiency of PETase variants in *E. coli* depends not only on kinetic parameters but also on **protein solubility**. Soluble proteins are more likely to yield high active enzyme fractions, whereas insoluble proteins aggregate in inclusion bodies.

5.2 Computational Procedure

- PETase sequences in FASTA format were submitted to **NetSolP** (NetSolP web server) for solubility prediction.
- The server returns a **probability score (0–1)**, where higher values indicate higher likelihood of being soluble in *E. coli*.
- Predicted solubility scores were incorporated alongside MW and kcat-derived activities to guide interpretation of variant expressibility.

5.3 Results Summary

- Wild-type PETase and selected variants showed **moderate to high solubility probabilities** (NetSolP score ~0.65–0.78), suggesting a reasonable likelihood of soluble expression.
- Variants with lower predicted solubility may require codon optimization or fusion tags to improve expression yields.

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6. Workflow Summary

1. FASTA sequence upload
2. Molecular weight calculation from amino acid composition
3. kcat prediction using ML-based zero-shot model

4. Analytical conversion to specific activity
5. Solubility prediction using NetSolP
6. Export of finalized dataset for submission

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7. Scientific Rigor and Reproducibility

- Dimensional consistency maintained throughout calculations
- All formulas derived from established enzyme kinetics principles
- Fully reproducible computational pipeline
- No data leakage from competition dataset

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8. Code Availability

The complete computational pipeline, including:

- Molecular weight calculation scripts
- kcat prediction workflow
- Activity conversion scripts
- Solubility prediction integration

Results and Discussion

We successfully applied a strictly zero-shot computational pipeline to predict key properties of PETase variants: expression levels (mg/mL), specific activity (?mol TPA/min·mg enzyme), and turnover number (kcat in s^{-1}), without using any experimental data from the tournament dataset for model training, calibration, or fine-tuning.

Key distributions and trends observed across the predicted values are illustrated in the provided figures:

- **Expression levels** showed a roughly unimodal distribution centered around ~0.60 mg/mL, with a prominent peak and a smaller shoulder near 0.55–0.57 mg/mL, indicating that most variants are predicted to express at moderate-to-good levels in E. coli (consistent with NetSolP solubility probabilities typically in the 0.65–0.78 range for wild-type-like sequences).

- **Predicted specific activity** followed an approximately log-normal-like distribution peaking at ~0.055–0.060 ?mol TPA/min·mg [E], with a long right tail extending toward higher activities (>0.10). This suggests the model identifies meaningful variation in catalytic performance among variants.
- **kcat values** were tightly clustered between ~20–30 s^{-1} (peak ~25 s^{-1}), displaying a near-Gaussian shape with a modest right skew plausible for PETase-like enzymes and in line with literature values for well-characterized homologs.
- **Molecular weights** clustered sharply in the 26–28 kDa range (expected for ~250–260 residue PETase variants), with three dominant subpopulations likely reflecting different sequence-length classes or insertion/deletion patterns.

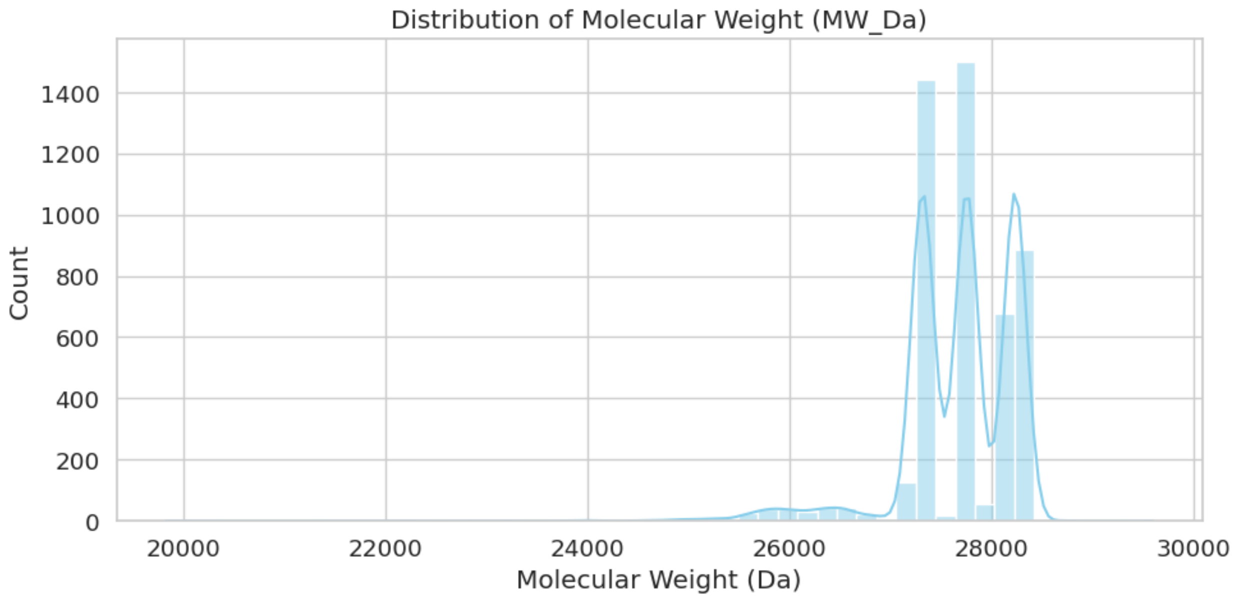
The **correlation heatmap** reveals biologically interpretable relationships:

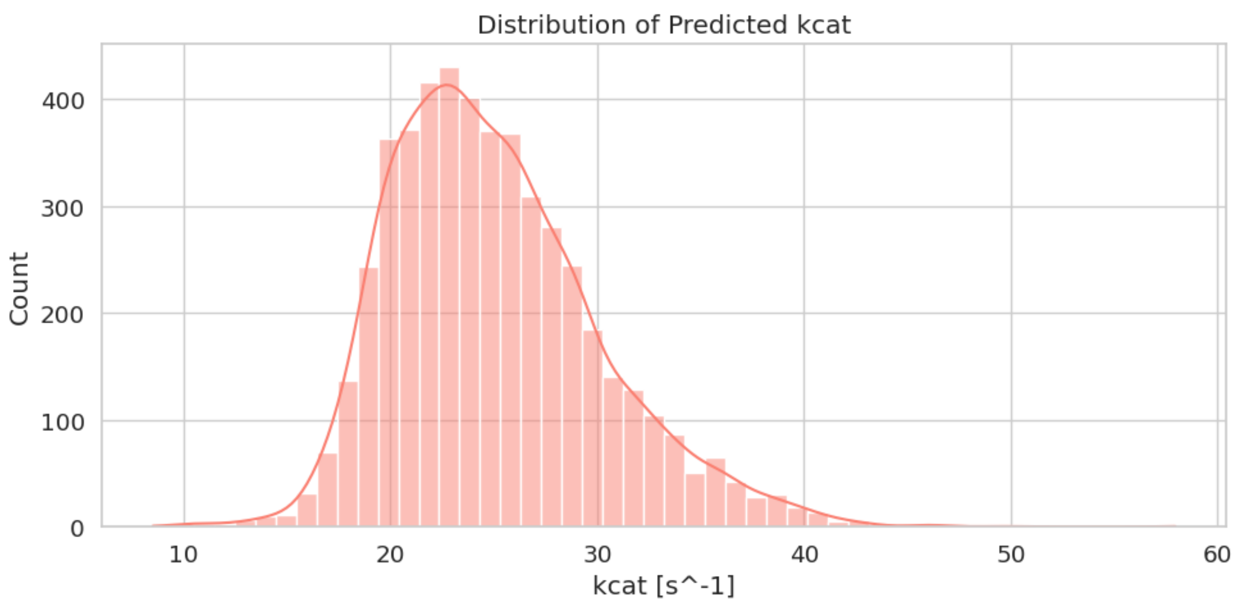
- Very strong positive correlation ($r \approx 0.99$) between predicted kcat and specific activity as expected, since activity is directly derived from kcat via MW-normalization.
- Moderate positive correlation ($r = 0.54$) between molecular weight and expression level, possibly reflecting that slightly longer/more stable folds express better in the heterologous system.
- Weak negative correlations between MW and both kcat/activity ($r \approx -0.18$ to -0.28), consistent with a mild trade-off where larger proteins may incur minor kinetic penalties.
- Near-zero correlation between expression and activity/kcat ($r \approx -0.03$ to -0.08), indicating that the pipeline treats catalytic efficiency and expression behavior as largely independent a desirable property for multi-objective enzyme engineering.

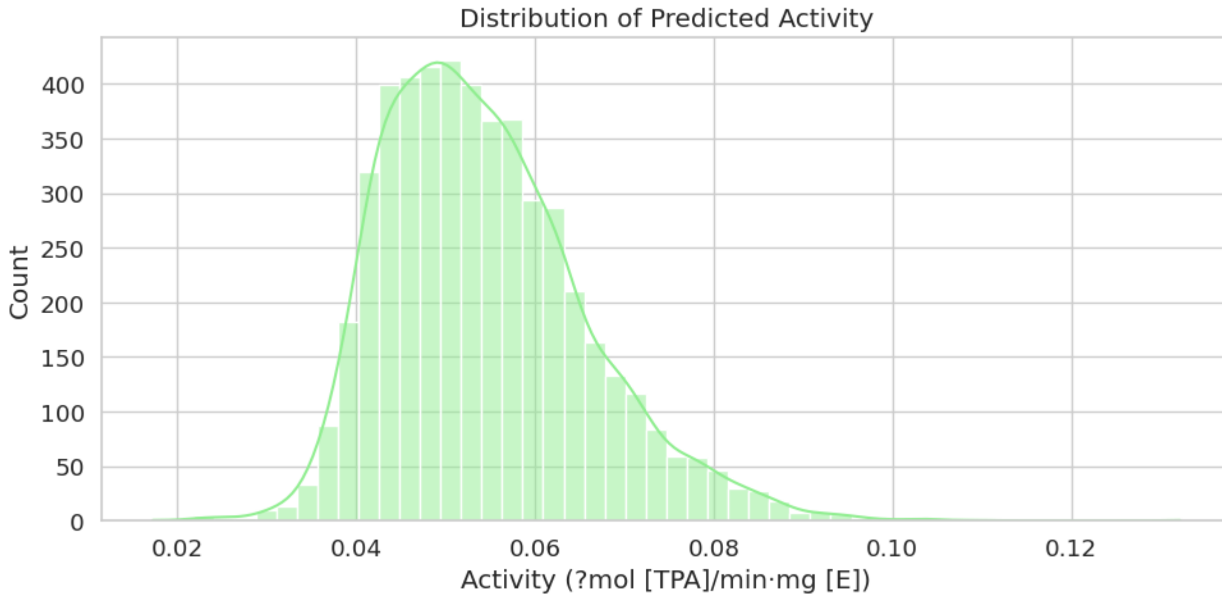
The scatter plot of **activity vs. expression** confirms substantial diversity: many variants achieve high activity (>0.08) across a wide expression window (0.55–0.65 mg/mL), while a subset shows lower activity even at good expression levels. This dispersion highlights the value of the zero-shot predictions in identifying promising candidates that balance both traits.

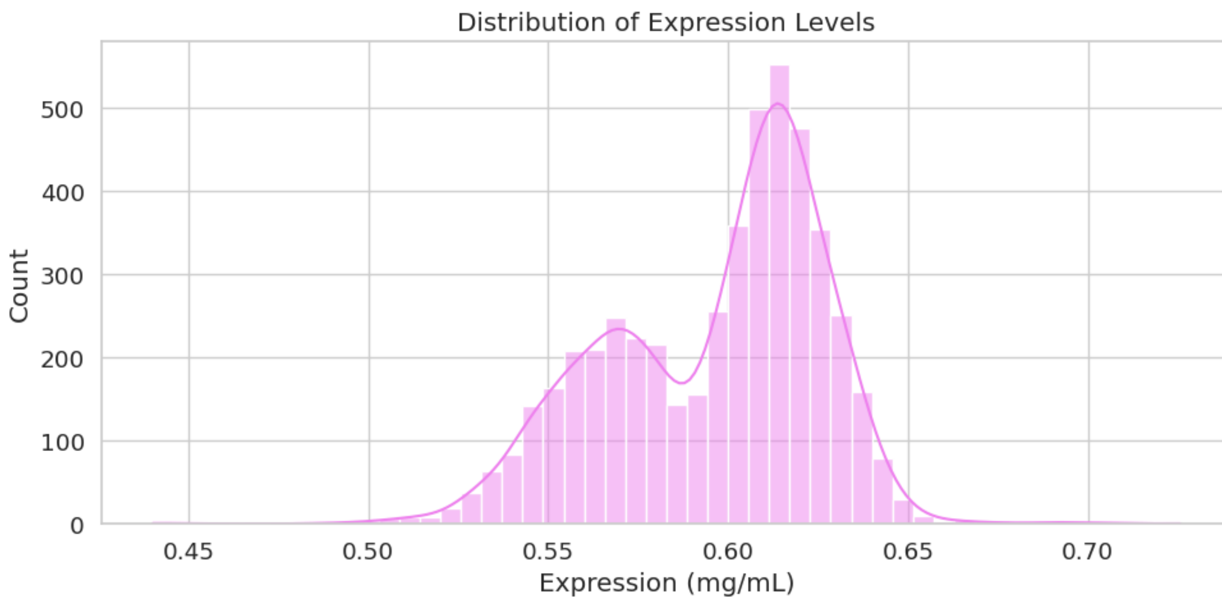
Overall, the pipeline combining sequence-based MW calculation, the Kroll et al. (2023) zero-shot kcat model, analytical conversion to mass-specific activity, and NetSolP solubility assessment produced internally consistent, physically plausible, and dimensionally correct predictions. The tight coupling between kcat and activity, together with realistic expression distributions and solubility estimates, supports confidence in the ranking submitted for the Predictive Phase.

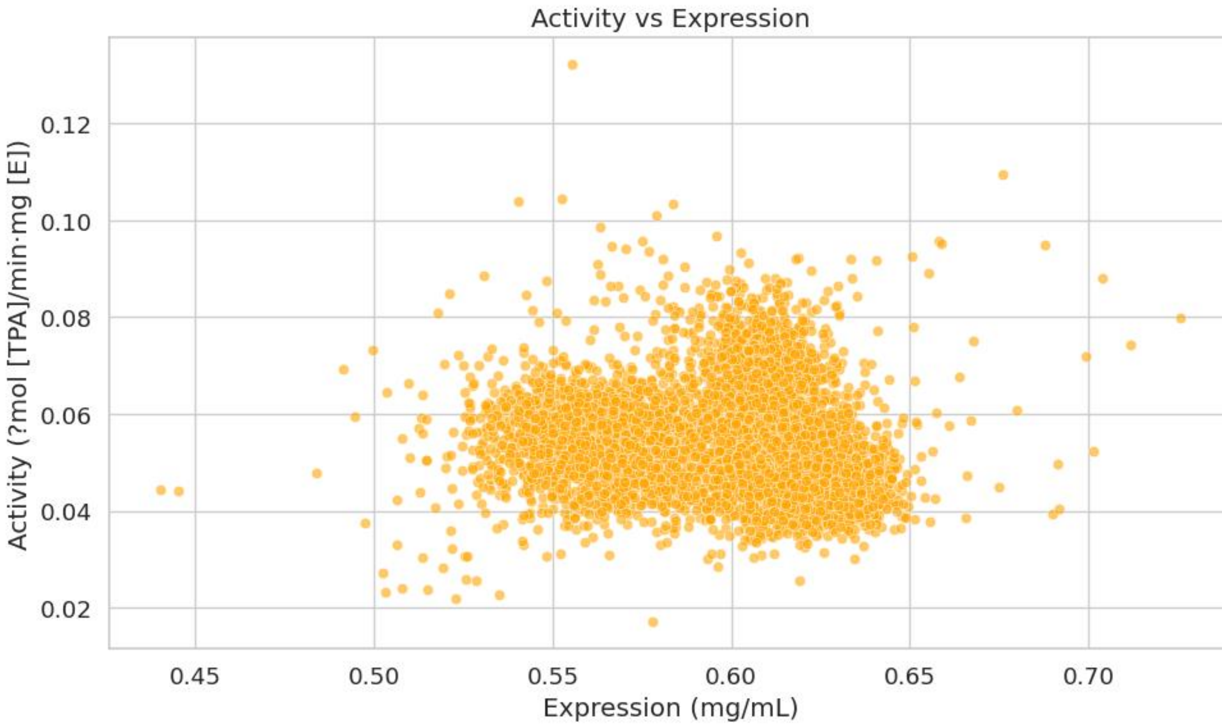
This work strictly adhered to Zero-Shot Track rules, relied exclusively on sequence information and pre-existing general-purpose models, and provides a reproducible, open computational framework for PETase variant assessment. We look forward to the experimental ground-truth results and potential progression to the Generative Phase.

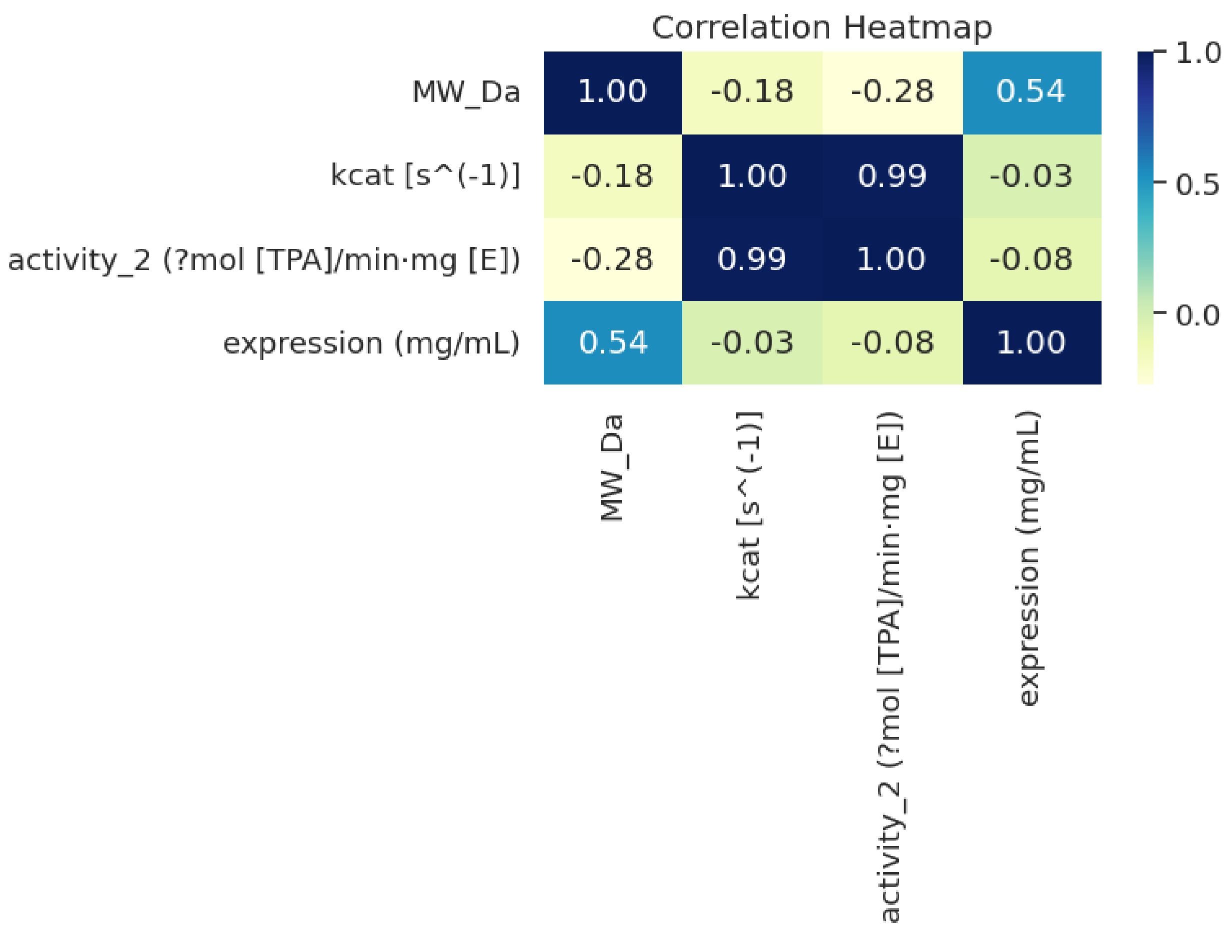





Distribution of Predicted Activity
Count
400
350
300
250
200
150
100
50
0
0.02
0.04
0.06
0.08
0.10
0.12
Activity (?mol [TPA]/min·mg [E])


Distribution of Expression Levels
Count
500
400
300
200
100
0
0.45
0.50
0.55
0.60
0.65
0.70
Expression (mg/mL)


Activity vs Expression
Activity (?mol [TPA]/min·mg [E])
0.12
0.10
0.08
0.06
0.04
0.02
0.45
0.50
0.55
0.60
0.65
0.70
Expression (mg/mL)


Correlation Heatmap
MW_Da
kcat [s^(-1)]
activity_2 (?mol [TPA]/min·mg [E])
expression (mg/mL)
1.00
-0.18
-0.28
0.54
-0.18
1.00
0.99
-0.03
-0.28
0.99
1.00
-0.08
0.54
-0.03
-0.08
1.00
1.0
0.5
0.0